GAN Dashboard

DATS 6303: Final Project

Jeongwon Yoo, Pratik Shetty, Josh Schechter

Generative AI models have improved considerably in recent years. For this final project, we explored several foundational generative AI models, with a specific focus on image generation. We selected variations of the Generative Adversarial Network (GAN) architecture, beginning with a simple DCGAN architecture, then developing a WGAN-GP implementation, and finally attempting to train a ProGAN model. After training was complete, we developed an interactive dashboard using Streamlit to display our results and demonstrate our understanding of each architecture. We used Frechet Inception Distance (FID) as our evaluation metric because it captures both the diversity of generated images and their distance from real images.

The dataset used for this project was a Kaggle dataset containing 140,000 real and fake images. This provided 70,000 real images for training the discriminator, as discussed later.

DCGAN served as our baseline GAN architecture and was the first model we implemented. Like a standard GAN, it consists of a generator and a discriminator trained against each other. The main difference is that DCGAN replaces fully connected image generation with a convolutional design, making it more suitable for image data. In our implementation, the generator starts from a 100-dimensional latent noise vector and upsamples it into a 64×64 RGB face image using transposed convolution layers, Batch Normalization, and ReLU, with a final Tanh output. The discriminator performs the reverse process by taking a 64×64 RGB image and downsampling it through convolution layers with LeakyReLU and Batch Normalization until it outputs a real/fake probability. This made DCGAN a strong baseline for understanding adversarial image generation before moving to WGAN-GP and ProGAN.

The WGAN architecture uses Earth Mover distance for the critic rather than the probability-based output used by DCGAN. The Earth Mover function outputs a raw scalar value that represents how far the generated image is from a real one. Because this approach does not rely on direct overlap between the real and sampled distributions, it allows for a smoother training curve.

However, there is an important caveat with the Earth Mover function: the critic must be 1-Lipschitz compliant. This means that the output of the function must not increase faster than a linear relationship with respect to the input. The original solution was weight clipping, which bounded the weights to help keep the outputs smooth. However, this approach was not ideal and led to poor results. This led to the introduction of a gradient penalty, which penalizes outputs that are too strong and adds more stability.

ProGAN takes a fundamentally different approach from DCGAN and WGAN-GP by progressively growing the generator and discriminator during training. Rather than training at the target resolution from the start, ProGAN begins at a very low resolution of 4×4 pixels and gradually doubles it (8×8, 16×16, 32×32) up to the target resolution of 64×64 or 128×128. This allows the model to first learn the global structure of a face before refining finer details such as textures and hair.

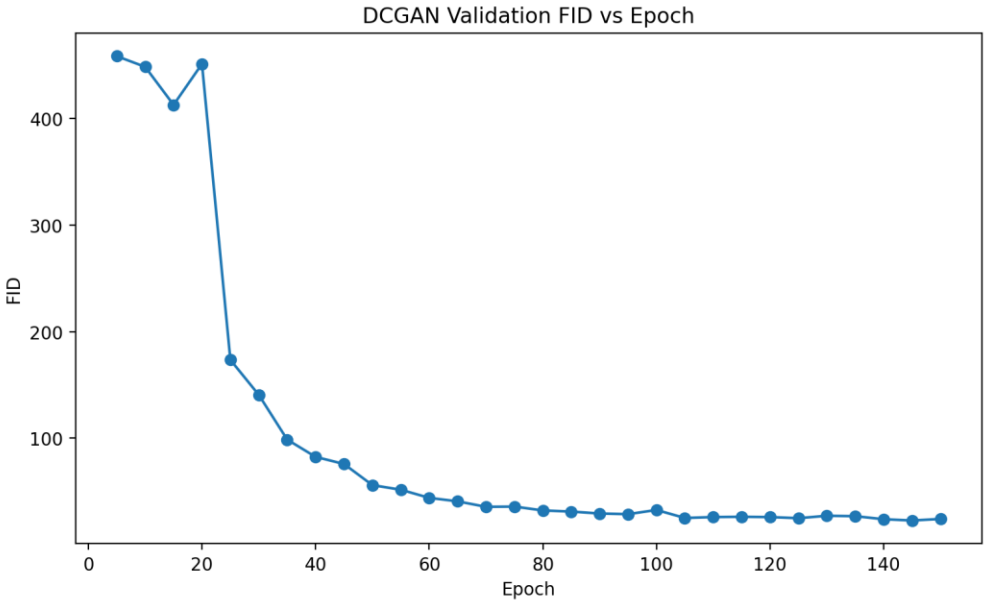
A key challenge with this approach is introducing new layers without disrupting what the network has already learned. ProGAN addresses this with a fade-in mechanism controlled by an alpha parameter that moves from 0 to 1. At alpha equals 0, the network relies entirely on the previous resolution. As alpha increases, the new layer is gradually blended in until it fully takes over at alpha equals 1. This smooth transition prevents sudden disruption to the already-learned features.

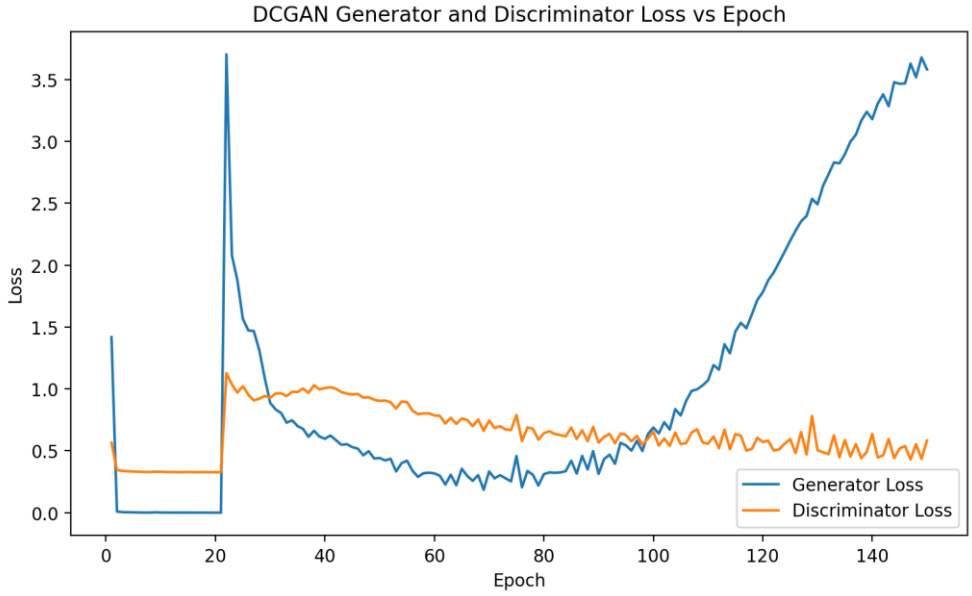
ProGAN also introduces three additional techniques to further stabilize training. Equalized learning rate scales the weights dynamically at runtime, ensuring that all layers train at the same effective speed. Pixelwise feature normalization prevents signal magnitudes from exploding in the generator. Finally, minibatch standard deviation adds diversity statistics to the discriminator, which discourages mode collapse, or the tendency for the generator to produce repetitive, similar images. Like WGAN-GP, ProGAN uses Wasserstein loss with a gradient penalty to enforce the Lipschitz constraint and maintain stable gradients throughout training.

The next phase was training. Each model was trained under the same preprocessing requirements.

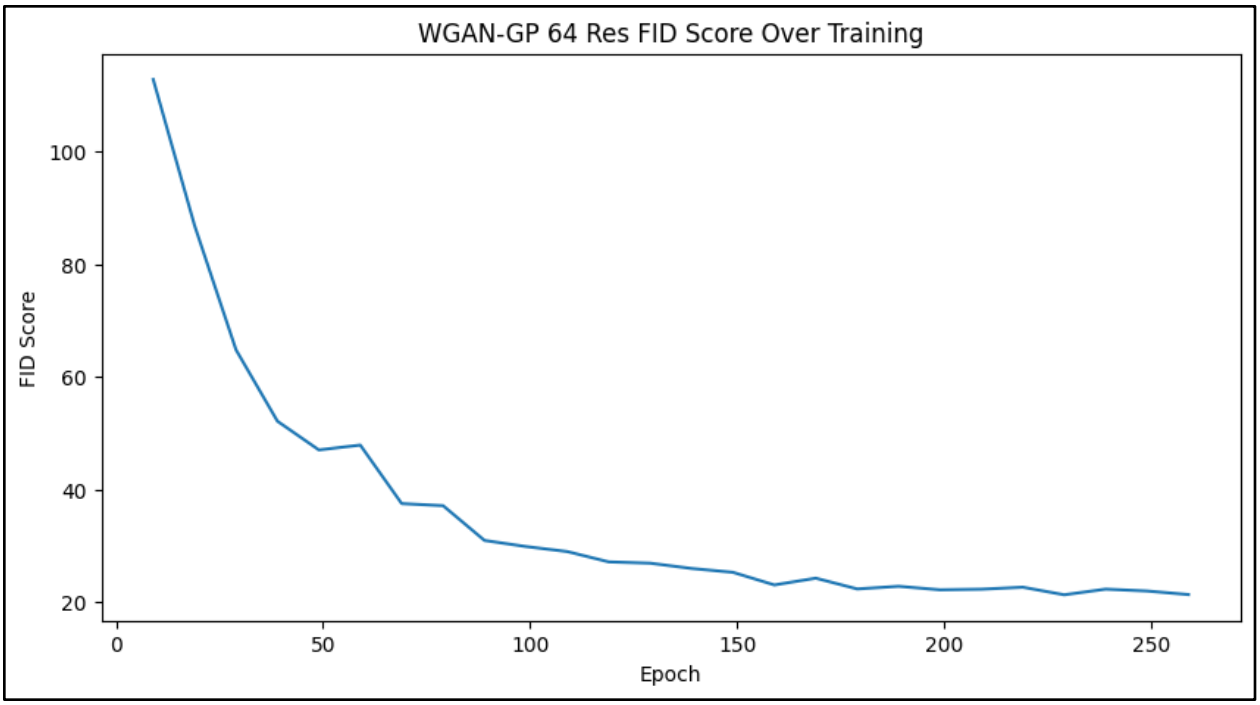
For training, DCGAN was used only at 64×64 resolution and served as our baseline comparison model. We trained it with a batch size of 128, 150 epochs, and feature maps of 128. The generator and discriminator used separate Adam learning rates, 1e-4 and 5e-5 respectively, to reduce instability. Training used binary cross-entropy loss, with real-label smoothing at 0.9 and annealed instance noise early in training to prevent the discriminator from becoming too strong too quickly. We evaluated the model using FID every five epochs and saved the checkpoint with the best validation FID, then reloaded that checkpoint at the end. DCGAN reached recognizable results quickly, but it was less stable than WGAN-GP; therefore, it was kept mainly as a baseline and reference model in the final comparison.

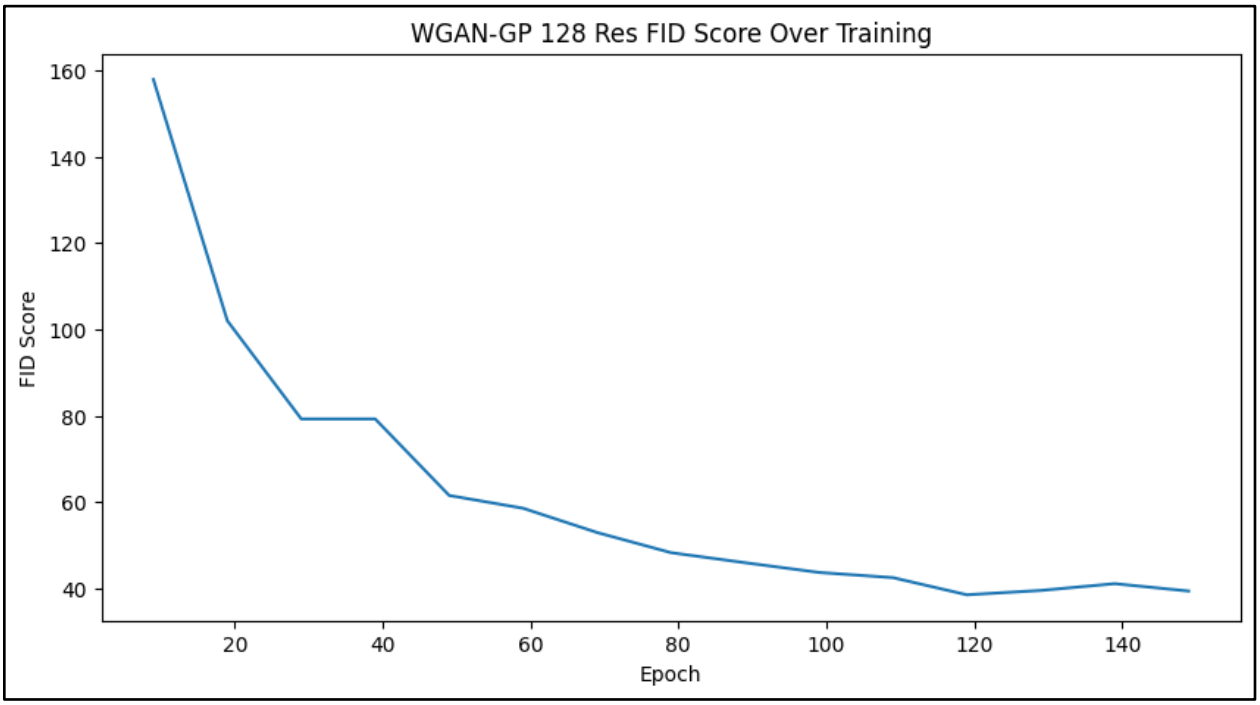
The following plots summarize DCGAN training behavior. The first plot shows validation FID over the training epochs, where lower values indicate that the generated images are becoming closer to the real data distribution. The second plot shows generator and discriminator loss across epochs, which illustrates the adversarial learning behavior between the two networks during training.



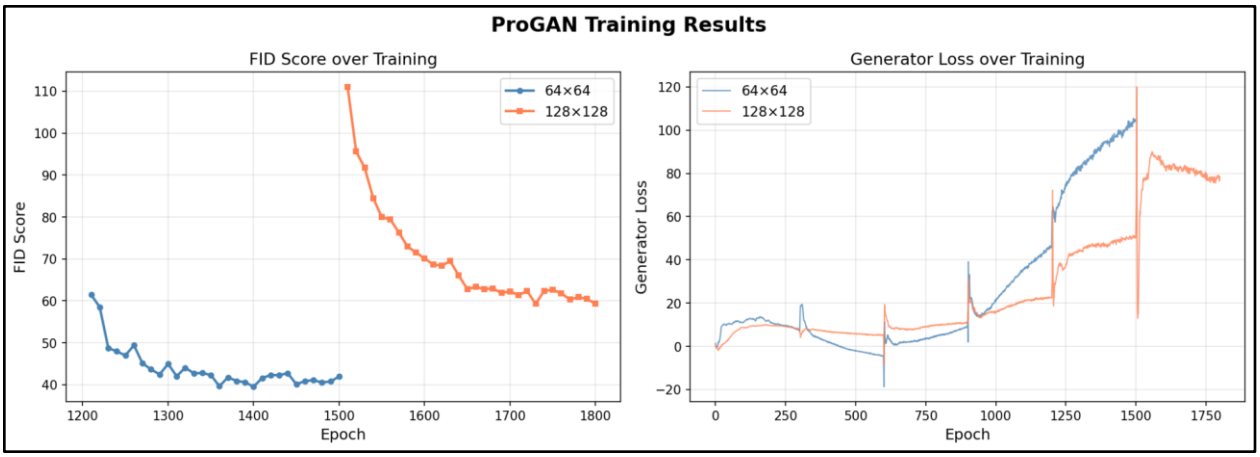


Training the WGAN-GP model was relatively straightforward. We ran tuning on a large search space once, then moved on to finer tuning and adjustments specific to each model. The 64×64 image-size run outperformed the 128×128 image-size run.





The following plots show the ProGAN results.



The left plot shows the FID score over training epochs for both resolutions. Both models demonstrate a consistent downward trend, indicating progressive improvement in image quality throughout training. The 64×64 model achieved a best FID of 39.5, while the 128×128 model reached a best FID of 59.3. The higher FID for the 128×128 model is expected, as generating higher-resolution images is inherently more challenging and requires more training to converge.

The right plot shows the generator loss over training epochs. Both models exhibit an initial period of instability during the early progressive growing steps, followed by gradual stabilization as training progresses. The spikes visible in the loss curves correspond to transitions between resolution steps, where new layers are introduced through the fade-in mechanism.

Real image example (64×64 pixels):

DCGAN image example (64×64 pixels):

WGAN-GP image example (64×64 pixels):

ProGAN image example (64×64 pixels):

ProGAN image example (128×128 pixels):

Based on our study, WGAN-GP performed the best for our dataset. In our experiment, WGAN-GP produced the strongest overall results. This was expected when compared with DCGAN, but it was surprising that it performed better than ProGAN, even at higher resolutions. There are a few possible explanations for this outcome. First, the image resolutions may not have been large enough for ProGAN to provide a clear benefit. Second, ProGAN is extremely difficult to train and tune, especially within a short period of time. With more time, it is very possible that ProGAN would have outperformed the other models at an image resolution of 128.

References

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EmilienDupont. wgan-gp GitHub repository. Consulted for reference while implementing WGAN-GP ideas; code was not copied or directly reused. https://github.com/EmilienDupont/wgan-gp